1. **Aim of the practical:** Make the created project interactive by using JavaScript**.**

2. **Objective:** To create a functional stopwatch that tracks elapsed time with start, stop, and reset functionalities using HTML, CSS, and JavaScript.

3. **Commands/Steps:**

CODE: Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Stopwatch</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="stopwatch">
    <h1 id="display">00:00:00</h1>
    <div class="buttons">
      <button id="startButton">Start</button>
      <button id="stopButton">Stop</button>
```

```
      <button id="resetButton">Reset</button>
    </div>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

Style.css

```
body {
    font-family: Arial, sans-serif;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    margin: 0;
    background-color: #f0f0f0;
  }

  .stopwatch {
    text-align: center;
    background: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.2);
  }

  h1 {
    font-size: 3rem;
    margin: 0 0 20px;
  }

  .buttons button {
```

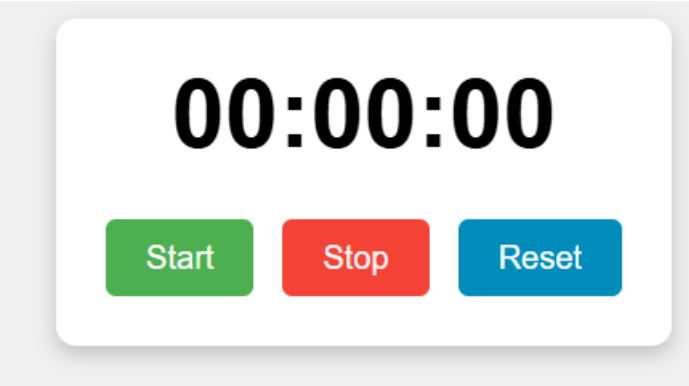
```
  font-size: 1rem;
  padding: 10px 20px;
  margin: 5px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
}

#startButton {
  background-color: #4caf50;
  color: white;
}

#stopButton {
  background-color: #f44336;
  color: white;
}

#resetButton {
  background-color: #008cba;
  color: white;
}

.buttons button:hover {
  opacity: 0.8;
}
```

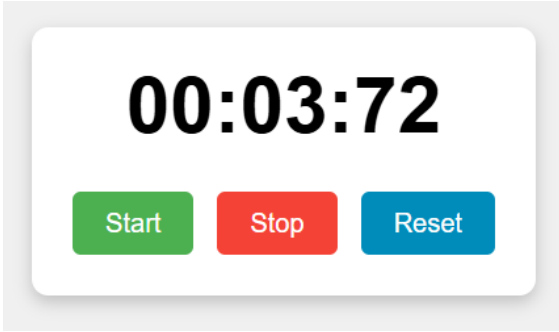
JavaScript:

```
// Select elements
const display = document.getElementById('display');
const startButton = document.getElementById('startButton');
const stopButton = document.getElementById('stopButton');
const resetButton = document.getElementById('resetButton');

// Variables to track time and interval
let startTime = 0;
let elapsedTime = 0;
let interval = null;

// Function to format time
function formatTime(time) {
  const milliseconds = Math.floor((time % 1000) / 10);
  const seconds = Math.floor((time / 1000) % 60);
  const minutes = Math.floor((time / (1000 * 60)) % 60);
  return `${String(minutes).padStart(2, '0')}:${String(seconds).padStart(2,
'0')}:${String(milliseconds).padStart(2, '0')}`;
}

// Start the stopwatch
startButton.addEventListener('click', () => {
  if (interval) return; // Prevent multiple intervals
  startTime = Date.now() - elapsedTime;
  interval = setInterval(() => {
    elapsedTime = Date.now() - startTime;
    display.textContent = formatTime(elapsedTime);
  }, 10);
});

// Stop the stopwatch
```

```
stopButton.addEventListener('click', () => {
  clearInterval(interval);
  interval = null;
});

// Reset the stopwatch
resetButton.addEventListener('click', () => {
  clearInterval(interval);
  interval = null;
  elapsedTime = 0;
  display.textContent = '00:00:00';
});
```

4. Result: STOP WATCH





5. Learning Outcomes (what I have learned):

- Understand how to use JavaScript timers (setInterval and clearInterval) for real-time updates.
- Learn to implement a responsive and interactive design using HTML, CSS, and JavaScript for practical applications.
- Gain experience in DOM manipulation to dynamically update the UI based on user interactions